

Fits a single machine

Exceeds a single machine

Run-time

PostGIS

Fastest in every single-machine cell

Apache Sedona

Lowest join run-time
from the medium tier

small tier: tied (either)

Cost

Small tier, by pattern

DuckDB **GeoPandas**

Large tier

PostGIS

Apache Sedona

Lowest cost on the
medium & large join

At the large join the single-node engines do not complete — a feasibility limit, not a question of speed.
No crossover was observed within 2-16 workers; any boundary beyond the tested range is an extrapolation.